Wireless Controlled 4-Wheel Drive Robot Car using ESP32

1. Abstract

This project presents the design and development of a wireless controlled 4-wheel drive robotic car using an ESP32 microcontroller. The system allows remote navigation of the robot through wireless communication, enabling efficient and flexible control. The robot uses a motor driver to control four DC motors and a buck converter to regulate voltage for stable operation.

2. Introduction

Robotics is an essential field in modern engineering that integrates electronics, programming, and mechanical design. This project demonstrates a practical implementation of a wireless robotic vehicle using ESP32.

The objective is to design a robot that can move in multiple directions using wireless commands.

3. Objectives

- To design a wireless controlled robotic car
- To interface ESP32 with motor driver
- To control multiple motors simultaneously
- To implement wireless communication

4. Components Description

ESP32: Main controller with WiFi/Bluetooth capabilities.

Motor Driver: Controls direction and speed of motors.

DC Motors: Provide movement to the robot.

Buck Converter: Steps down voltage to safe level.

Wireless Module: Enables communication.

Chassis: Physical structure of robot.

5. System Architecture

The system consists of a transmitter (controller) and receiver (robot).

The controller sends commands wirelessly to ESP32.

ESP32 processes signals and drives motors using motor driver.

6. Working Principle

When a command is sent from controller, ESP32 receives it and executes corresponding motor action.

Different combinations of motor directions allow movement in all directions.

7. Motor Logic

Forward: All motors rotate forward.

Backward: All motors rotate backward.

Left: Left motors reverse, right motors forward.

Right: Left motors forward, right motors reverse.

Stop: All motors off.

8. Power Supply

Battery supplies power to system.

Buck converter regulates voltage to 5V.

Common ground ensures stable operation.

9. Advantages

• Easy to control

• Low cost

• Expandable design

• Wireless flexibility

10. Applications

• Surveillance robots

- Industrial automation
- Remote vehicles
- Educational projects

11. Future Enhancements

- Obstacle avoidance
- Camera integration
- Voice control system
- AI-based navigation

12. Conclusion

This project successfully demonstrates wireless robot control using ESP32. It serves as a foundation for advanced robotic systems.